

Uzbekistan OS

Database Design Specification (DDS)

Version 1.0

1. Purpose

Defines the database architecture for the Uzbekistan OS MVP, including schemas, tables, relationships, constraints, indexes, vector storage, workflows, conversations, AI observability, ingestion, analytics, and migrations.

2. Design Goals

Support traceable AI answers, versioned knowledge, hybrid retrieval, multilingual content, structured workflows, country portability, safe publishing, operational observability, and MVP simplicity.

3. Architecture

Use PostgreSQL as the system of record with pgvector, JSONB, UUID primary keys, UTC timestamps, soft deletion where appropriate, immutable knowledge versions, and explicit publication states.

4. PostgreSQL Schemas

public, identity, geography, knowledge, workflow, conversation, ai, ingestion, analytics, and audit.

5. Core Extensions

pgcrypto, vector, pg_trgm, unaccent, and citext.

6. Geography & Localization

Countries, regions, cities, languages, and nationalities designed for future multi-country expansion.

7. Identity

Users, authentication identities, guest sessions, profiles, and user preferences.

8. Knowledge Layer

Domains, topics, tags, source organizations, sources, snapshots, documents, document versions, document sources, applicability rules, chunks, embeddings, and translation support.

9. Workflow Layer

Workflow definitions, steps, dependencies, user workflow progress, checklist templates, and personalized checklists.

10. Conversation Layer

Conversations, messages, citations, and user feedback.

11. AI Layer

Model registry, embedding models, AI requests, retrieval runs, retrieval results, and prompt version management.

12. Ingestion

Crawl jobs, extraction runs, change events, and review workflows for knowledge validation.

13. Analytics & Audit

Product analytics events and immutable audit logs.

14. Index Strategy

Indexes optimized for identity, retrieval, knowledge, conversations, AI logs, and ingestion queues.

15. Retrieval Rules

Only verified, published, applicable, effective, and supported knowledge is eligible for semantic retrieval.

16. Security

Row-level security (where applicable), least-privilege access, data retention, PII minimization, encrypted backups, and migration strategy.

17. Seed Data

Countries, languages, domains, topics, source organizations, model configuration, prompt versions, and development admin.

18. MVP Implementation Order

Create schemas and extensions, geography, knowledge, retrieval, identity, conversations, AI logging, workflows, ingestion, analytics, audit, then views and permissions.

19. Future Expansion

Support for multi-country deployments, knowledge graphs, OCR, appointments, government integrations, subscriptions, and AI agents without redesigning the core schema.

20. Definition of Done

All migrations succeed, relationships enforced, retrieval operational, conversations persisted, AI logging functional, workflows executable, backups restorable, and integration tests passing.